

Rectilinear Motion

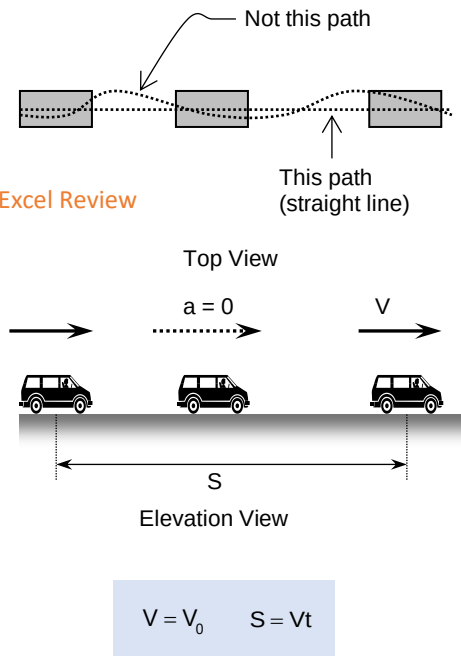
Rectilinear translation is a motion in a plane or one dimension.

Types of Rectilinear Translations:

1. Horizontal translation
2. Vertical translation
3. Curvilinear translation

Horizontal Translation:

Horizontal translation (motion) is a straight motion along a horizontal plane. If acceleration is zero ($a = 0$), the velocity of the body remains constant throughout the motion, thus:



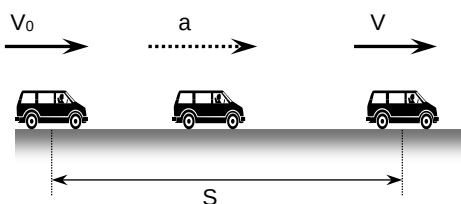
where:

V_0 = initial velocity (time = 0)

V = final velocity (time = t)

S = distance traveled after elapsed time t

If acceleration is constant, the initial velocity, V_0 is not equal to the final velocity, V .



The following formulas will be used:

$$S = V_0 t + \frac{1}{2} a t^2 \quad V = V_0 + a t$$

$$V^2 = V_0^2 + 2 a S$$

where:

V_0 = initial velocity (time = 0)

V = final velocity (time = t)

S = distance traveled after time, t

a = acceleration

Vertical Translation:

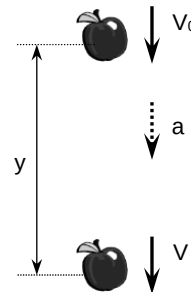
Vertical translation (motion) is a straight motion along a vertical plane. In this type of motion, acceleration cannot be equal to zero since the velocity of a body moving in vertical motion, increases as it goes down and decreases as it goes up.

The following formulas will be used:

$$V^2 = V_0^2 + 2 a y$$

$$V = V_0 + a t$$

$$y = V_0 t + \frac{1}{2} a t^2$$



where:

V_0 = initial velocity (time = 0)

V = final velocity (time = t)

S = distance traveled after time t

a = acceleration

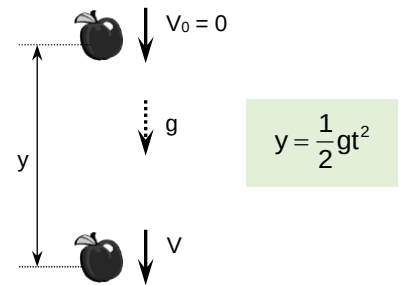
If the acceleration of the body is not given, then it is presumed that the acceleration to be used is the acceleration due to gravity (gravity on earth).

$$g = 9.81 \text{ m/s}^2 \text{ or } g = 32.2 \text{ ft/s}^2$$

Free-Falling bodies

A body is said to be free falling if the initial vertical velocity is zero. The acceleration of a free-falling body is due to gravity which is equal to 9.81 m/s^2 or 32.2 ft/s^2 .

Excel Review

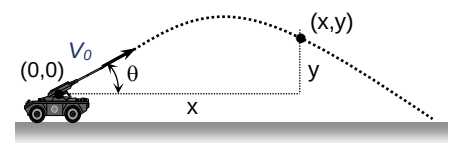


Curvilinear Motion

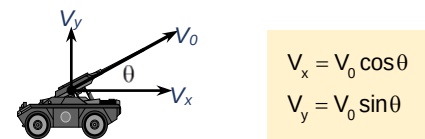
Curvilinear translation (motion) is a motion along a curve path either on the vertical plane or on horizontal plane. This type of motion includes a trajectory or projectile, which is a curve path on the vertical plane and rotation on horizontal plane.

Excel Review

A **projectile** is a body which after being given an initial velocity with an initial angle of release is allowed to travel under the action of gravity only. A projectile is a trajectory which is a graph of a **parabola**.



Let: V_x and V_y be the horizontal component and the vertical component, respectively of the initial velocity, V_0 .



Since the horizontal component of the velocity is constant throughout the projectile, then the value of x is calculated as follows:

$$x = V_0 \cos \theta t$$

So the time of flight, t is:

$$t = \frac{x}{V_0 \cos \theta}$$